

Magnetism

Questions with Fully Worked Answers · FormulaVerse PYQ Bank

Q1. Two identical long current carrying wires are bent into shapes shown in the figure — shape (I): two straight semi-infinite wires tangent to a semicircular arc of radius r at centre P ; shape (II): one straight semi-infinite wire with a semicircular arc of radius r at centre Q . If the magnitudes of magnetic field at P and Q are B_1 and B_2 respectively, find B_1/B_2 .

NTA B.Tech · 2 Apr 2026, Shift 1 (Q39)

- (1) $(2+\pi)/(1+\pi)$
- (2) $(1+\pi)/(1-\pi)$
- (3) $(2+\pi)/(1-\pi)$
- (4) $(1+\pi)/(2-\pi)$

✓ **Answer: (1) $(2+\pi)/(1+\pi)$**

Solution:

Semicircular arc field at its centre = $\mu_0 I/(4r)$ in both cases.

Each semi-infinite straight wire tangent to the centre contributes $\mu_0 I/(4\pi r)$.

Shape (I) has 2 such straight segments: $B_1 = \mu_0 I/(4r) + 2 \times \mu_0 I/(4\pi r) = (\mu_0 I/r) \times (\pi+2)/(4\pi)$.

Shape (II) has 1 straight segment: $B_2 = \mu_0 I/(4r) + \mu_0 I/(4\pi r) = (\mu_0 I/r) \times (\pi+1)/(4\pi)$.

Ratio $B_1/B_2 = (\pi+2)/(\pi+1)$.

Q2. $1 \mu\text{C}$ charge moving with velocity $\mathbf{v} = (\hat{i} - 2\hat{j} + 3\hat{k})$ m/s in a region of magnetic field $\mathbf{B} = (2\hat{i} + 3\hat{j} - 5\hat{k})$ T. The magnitude of force acting on it is $\sqrt{\alpha} \times 10^{-6}$ N. The value of α is ____.

NTA B.Tech · 2 Apr 2026, Shift 1 (Q48)

✓ **Answer: 171**

Solution:

$$\mathbf{F} = q(\mathbf{v} \times \mathbf{B}).$$

$$\begin{aligned}\mathbf{v} \times \mathbf{B} &= (v_y B_z - v_z B_y, v_z B_x - v_x B_z, v_x B_y - v_y B_x) \\ &= ((-2)(-5) - (3)(3), (3)(2) - (1)(-5), (1)(3) - (-2)(2)) = (1, 11, 7).\end{aligned}$$

$$|\mathbf{v} \times \mathbf{B}| = \sqrt{1+121+49} = \sqrt{171}.$$

$$\mathbf{F} = q \times \sqrt{171} = 10^{-6} \times \sqrt{171} \text{ N. So } \alpha = 171.$$

Source: NTA B.Tech — 2 April 2026, Shift 1 (official NTA release, physics section) · formulaverse.in